

## 마이크로/나노공학 (공) (MECH402-01)

|      |                                                |
|------|------------------------------------------------|
| 학기   | 2026년 1학기                                      |
| 담당교수 | 이재홍                                            |
| 이수구분 | 교선평수                                           |
| 학점   | 3.0                                            |
| 강의시간 | 화13:00-14:30(E7 - 229), 목13:00-14:30(E7 - 229) |
| 강의실  | E5 611 (교수실)                                   |
| 소속   | 기초학부                                           |

### 교과개요

This course introduces overall history and development of microelectromechanical system (MEMS). Main components including sensors and actuators and microfabrication process for the MEMS will be mainly provided in addition to relevant theoretical basics and applications.

### 교과목표

This course provides a basic understanding regarding MEMS based on comprising components, microfabrication process, and applications. This course will be helpful for students who is eager to study micro-electronics, microfabrications, sensors, actuators, and various electronic devices.

### 주차별 강의계획

- 1주차:Introduction to MEMS
- 2주차:First-pass introduction to microfabrication
- 3주차:First-pass introduction to microfabrication
- 4주차:Review of Essential Electrical and Mechanical Concepts
- 5주차:Electrostatic Sensing and Actuators
- 6주차:Thermal Sensing and Actuation
- 7주차:Midterm exam
- 8주차:Piezoresistive Sensors
- 9주차:Piezoelectric Sensing and Actuation
- 10주차:Magnetic Actuation
- 11주차:Summary of Sensing and Actuation Methods
- 12주차:Bulk Micromachining and Silicon Anisotropic Etching
- 13주차:Surface Micromachining
- 14주차:Polymer MEMS
- 15주차:Biosensors
- 16주차:Final exam

### 학습윤리

Comply with University education policy

### 교과정책(방침)

- To come to class on time
- Not to use cell phones during class

- To turn on the screen during class to check attendance (Online-class)